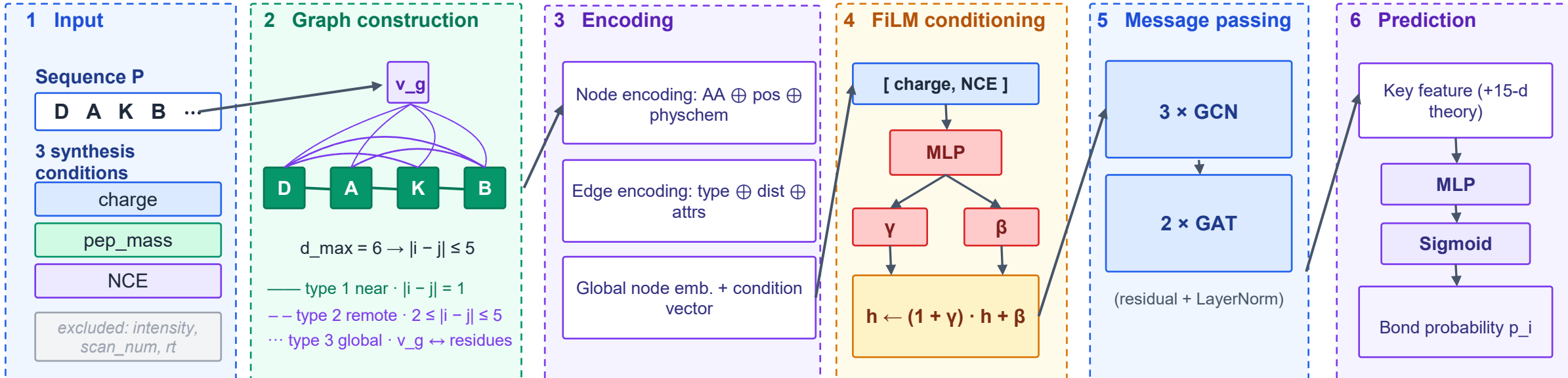

1 Input
Sequence P
D A K B ···
3 synthesis conditions
charge
pep_mass
NCE
excluded: intensity, scan_num, rt
2 Graph construction
v_g
D
A
K
B
d_max = 6 → |i − j| ≤ 5
type 1 near · |i − j| = 1
type 2 remote · 2 ≤ |i − j| ≤ 5
type 3 global · v_g ↔ residues
3 Encoding
Node encoding: AA ⊕ pos ⊕ physchem
Edge encoding: type ⊕ dist ⊕ attrs
Global node emb. + condition vector
4 FiLM conditioning
[charge, NCE]
MLP
γ
β
h ← (1 + γ) · h + β
5 Message passing
3 × GCN
2 × GAT
(residual + LayerNorm)
6 Prediction
Key feature (+15-d theory)
MLP
Sigmoid
Bond probability p_i